

MAT 362 HW03

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Problem. 2.2.1. Show that the cylinder $\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = 1\}$ is a regular surface, and find parametrizations whose coordinate neighborhoods cover it.

Solution. Consider the four maps $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4 : (-1, 1) \times \mathbb{R} \rightarrow \mathbb{R}^3$ defined as follows:

$$\mathbf{x}_1(x, z) = (x, \sqrt{1 - x^2}, z)$$

$$\mathbf{x}_2(x, z) = (x, -\sqrt{1 - x^2}, z)$$

$$\mathbf{x}_3(y, z) = (\sqrt{1 - y^2}, y, z)$$

$$\mathbf{x}_4(y, z) = (-\sqrt{1 - y^2}, y, z).$$

The union of the images of the $\mathbf{x}_i$ is the cylinder itself, the images of $\mathbf{x}_1$ and $\mathbf{x}_2$ together cover all points except the ones with a y -coordinate of zero, while $\mathbf{x}_3$ and $\mathbf{x}_4$ cover the points on the cylinder with nonzero x -coordinate.

We will verify that $\mathbf{x}_1$ is a parametrization of its image $\mathbf{x}_1(\mathbb{R}^2)$; the other verifications are analogous, so this will prove that the cylinder is a regular surface.

First, it's clear that all the component functions of $\mathbf{x}_1$ are smooth; in particular $\sqrt{1 - x^2}$ is a composition of smooth functions so it is smooth.

Second, $\mathbf{x}_1$ is a homeomorphism between $\mathbb{R}^2$ and its image $\mathbf{x}_1(\mathbb{R}^2)$, since it is clearly continuous and it has a continuous inverse in the form of a projection onto the xz -plane.

Third, its differential map is the matrix

$$\begin{pmatrix} 1 & 0 \\ * & 0 \\ 0 & 1 \end{pmatrix}$$

where $*$ is some function of x that we don't care about; this matrix has rank two because the rows $(1, 0)$ and $(0, 1)$ are unconditionally independent of one another. $\square$

Problem. 2.2.3. Show that the two-sheeted cone with its vertex at the origin (that is, the set $S = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 - z^2 = 0\}$) is not a regular surface.

Solution. Suppose that $\mathbf{x} : U \rightarrow V \cap S$ is a parametrization that defines a coordinate neighborhood $V \cap S$ of 0. Then, observe that $(V \cap S) \setminus \{0\}$ is disconnected because $S \setminus \{0\}$ is disconnected. We show that this implies S cannot be regular.

Claim — If U is an open set in $\mathbb{R}^3$, then it has no cut points; that is, there are no points $x \in U$ such that $U \setminus \{x\}$ is disconnected.

Proof. Suppose U has a cut point x and let $U' = U \setminus \{x\}$. Then, there exist nonempty open sets $A, B \subseteq \mathbb{R}^3$ that do not intersect in U' and whose union contains U' . Let $B_\varepsilon(x) \subseteq U$ be an open ball centered at x with radius ε contained inside U . It is easy to show that $B_\varepsilon(x) \setminus \{x\}$ is connected; in fact, it is path connected, for the ball (with x) is convex, and so two points in the ball without x can be connected by a segment, or, if the segment passes through x , then by a path made by joining two segments, with the intermediary endpoint not being collinear with the endpoints of the path.

Since $B_\varepsilon(x) \setminus \{x\}$ is connected, it must lie entirely in one of A or B — say WLOG A — else A and B form a separation of $B_\varepsilon(x) \setminus \{x\}$, contradicting its connectedness. But then the set $A' = A \cup \{x\}$ is open because it is the union of A with $B_\varepsilon(x)$ and doesn't intersect B inside U , so A and B exhibit that U is disconnected, which is a contradiction. $\square$

Since U and $V \cap S$ are homeomorphic, a cut point of U maps to a cut point of $V \cap S$ and vice versa — but U has no cut points, so $V \cap S$ must have no cut points. This is false in our situation here, so S cannot be regular. $\square$

Problem. 2.2.7. Let $f(x, y, z) = (x + y + z - 1)^2$.

- (a) Locate the critical points and critical values of f .
- (b) For what values of c is the set $f(x, y, z) = c$ a regular surface?

Solution. The differential of f is

$$df_{(x,y,z)} = 2(x + y + z - 1) \cdot (1, 1, 1).$$

Thus (x, y, z) is a critical point iff $x + y + z - 1 = 0$. The corresponding critical value is just $f(x, y, z) = 0$.

Since $f^{-1}(c)$ is a regular surface whenever c is a regular value, any $c \neq 0$ makes the set a regular surface.

We then have to check whether $f(x, y, z) = 0$ is a regular surface. This is trivial; the set $f^{-1}(0)$ is $\{(x, y, z) \in \mathbb{R}^3 : x + y + z = 1\}$ which is a plane and thus regular, for example with the parametrization $(u, v) \mapsto (u, v, 1 - u - v)$. Clearly this is infinitely differentiable and a homeomorphism between $\mathbb{R}^2$ and the plane, and the derivative

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{pmatrix}$$

is obviously injective. $\square$

Problem. 2.2.16. One way to define a system of coordinates for the sphere S^2 , given by $x^2 + y^2 + (z - 1)^2 = 1$, is to consider the so-called **stereographic projection** $\pi: S \setminus \{N\} \rightarrow \mathbb{R}^2$ which carries a point $p \in S^2$ that is not equal to the north pole $N = (0, 0, 2)$ to the intersection of the xy -plane with the line through N and p . Let $(u, v) = \pi(x, y, z)$ where $(x, y, z) \in S^2 \setminus \{N\}$.

- (a) Show that $\pi^{-1}: \mathbb{R}^2 \rightarrow S^2$ is given by

$$\pi^{-1}(u, v) = \begin{pmatrix} \frac{4u}{u^2 + v^2 + 4} \\ \frac{4v}{u^2 + v^2 + 4} \\ \frac{2(u^2 + v^2)}{u^2 + v^2 + 4} \end{pmatrix}.$$

- (b) Show that it is possible, using stereographic projection, to cover the sphere with two coordinate neighborhoods.

Solution. (a) Let $O = (0, 0, 0)$ be the origin, $A = (u, v, 0)$ be the point in the xy -plane, and $B = \pi^{-1}(u, v)$ be the point we are trying to find, namely the second intersection of line $\overline{NA}$ with the sphere. Then, since $\overline{ON}$ is a diameter of the sphere, $\angle NBO = \frac{\pi}{2}$, so we are trying to compute the foot of the altitude from O to line $\overline{NA}$. The vector $\overrightarrow{OB}$ can then simply be computed as

$$\begin{aligned} \overrightarrow{ON} + \overrightarrow{NB} &= (0, 0, 2) + \frac{\overrightarrow{NO} \cdot \overrightarrow{NA}}{\|\overrightarrow{NA}\|^2} \cdot \overrightarrow{NA} \\ &= (0, 0, 2) + \frac{(0, 0, -2) \cdot (u, v, -2)}{\|(u, v, -2)\|^2} \cdot (u, v, -2) \\ &= (0, 0, 2) + \frac{4}{u^2 + v^2 + 4} \cdot (u, v, -2) \\ &= \frac{1}{u^2 + v^2 + 4} (4u, 4v, -8 + 2(u^2 + v^2 + 4)) \\ &= \frac{1}{u^2 + v^2 + 4} (4u, 4v, 2(u^2 + v^2)). \end{aligned}$$

(b) Let $\pi_1 \equiv \pi$ be the stereographic projection of $S^2 \setminus N$ onto the xy -plane and let π_2 be the stereographic projection of $S^2 \setminus \{O\}$ onto the plane $z = 2$, where $O = (0, 0, 0)$ is the south pole. To be more precise π_2 projects the sphere onto $z = 2$ and then forgets the z -coordinate/projects $z = 2$ to the xy -plane. I claim that these two maps exhibit a covering of S^2 with the two coordinate neighborhoods $S^2 \setminus \{N\}$ and $S^2 \setminus \{O\}$.

We can see clearly that $\pi_1^{-1}: \mathbb{R}^2 \rightarrow S^2 \setminus \{N\}$ and $\pi_2: \mathbb{R}^2 \rightarrow S^2 \setminus \{O\}$ are differentiable of all orders (the only possible obstruction would be creating a denominator of zero somehow, but the denominators of the rational functions would just be powers of $u^2 + v^2 + 4$, which are always nonnegative), so this verifies condition (1).

Note now that π_1^{-1} and π_2^{-1} are bijections (since they are invertible, having inverses π_1, π_2 , by part (a)) from an open subset of $\mathbb{R}^2$ to an open subset S^2 (where “open” is with respect to the subspace S^2 , i.e. the intersection of an open set in $\mathbb{R}^2$ with S^2). To check condition (2) we need to also check that π_1, π_2 are continuous. We will just check it for π_1 by directly computing $\pi_1(x, y, z)$. If we are given

$$\pi^{-1}(u, v) = (x, y, z),$$

then we can compute

$$z = \frac{2(u^2 + v^2)}{u^2 + v^2 + 4} \iff u^2 + v^2 = \frac{4z}{2 - z}$$

(noting that $2 - z \neq 0$ because $z < 2$) and then

$$x = \frac{4u}{4z/(2 - z) + 4} = \frac{4u}{8/(2 - z)} = \frac{u(2 - z)}{2} \iff u = \frac{2x}{2 - z}$$

and similarly $v = \frac{2y}{2 - z}$. These are clearly continuous in x, y, z , so π_1 is continuous. A similar proof follows for π_2 (or one can apply the change of coordinates $z \mapsto 2 - z$ to reflect the stereographic projection over $z = 1$, swapping O and N as well as the planes that are being mapped to). We are hence done verifying condition (2).

To finish proving that these are coordinate neighborhoods we need to check that the derivatives of π_1^{-1} and π_2^{-1} are injective. Again we check this only in the case of π_1^{-1} : first we compute

$$d\pi_1^{-1}(u, v) = \frac{1}{(u^2 + v^2 + 4)^2} \begin{pmatrix} 4(-u^2 + v^2 + 4) & -8uv \\ -8uv & 4(u^2 - v^2 + 4) \\ 16u & 16v \end{pmatrix}.$$

To check that this linear map is one-to-one, we need the three equations

$$\begin{aligned} 4(-u^2 + v^2 + 4) &= -8\lambda uv \\ -8uv &= 4\lambda(u^2 - v^2 + 4) \\ 16u &= 16\lambda v \end{aligned}$$

to simultaneously hold for some real $\lambda \neq 0$, noting that neither of the two columns can be all zero at the same time, since either u or v would have to be zero but then $4(-u^2 + v^2 + 4)$ or $4(u^2 - v^2 + 4)$ respectively would be nonzero. This also means we can assume $u, v \neq 0$. We can now get $\lambda = \frac{u}{v}$ so that

$$\begin{aligned} 4(-u^2 + v^2 + 4) &= -8u^2 \\ 4(u^2 - v^2 + 4) &= -8v^2. \end{aligned}$$

In fact both equations are equivalent to $4u^2 + 4v^2 = -16$, which is impossible because squares are nonnegative. This checks off condition (3) for π_1^{-1} , and by an analogous argument or by symmetry, we can conclude that it holds for π_2^{-1} as well, thus finishing the problem. $\square$

Problem. 2.3.1. Let S^2 be the unit sphere and let $A: S^2 \rightarrow S^2$ be the **antipodal map** $A(x, y, z) = -(x, y, z)$. Prove that A is a diffeomorphism.

Solution. Since A is its own inverse, we only need to show that A is differentiable on all of S^2 . By definition, we wish to show that for all $x \in A$ and for any two parametrizations $\mathbf{x}_1: U_1 \rightarrow S^2$, $\mathbf{x}_2: U_2 \rightarrow S^2$ such that $x \in \mathbf{x}_1(U_1) \cap \mathbf{x}_2(U_2)$, the composition

$$\mathbf{x}_2^{-1} \circ A \circ \mathbf{x}_1$$

is differentiable at $\mathbf{x}_1^{-1}(x)$ in the sense that a map from $\mathbb{R}^2$ to $\mathbb{R}^2$ is differentiable. This is immediate in light of the fact that $-v \mapsto v$ is a differentiable map on all of $\mathbb{R}^3$, and compositions of differentiable functions are differentiable. $\square$

Problem. 2.3.2. Let $S \subseteq \mathbb{R}^3$ be a regular surface and let $\pi: S \rightarrow \mathbb{R}^2$ be the map that takes each $p \in S$ into its orthogonal projection over $\mathbb{R}^2 = \{(x, y, z) \in \mathbb{R}^3 : z = 0\}$. Is π differentiable?

Solution. We observe that $\pi(x, y, z) = (x, y, 0)$ is differentiable when treated as a function from $\mathbb{R}^3$ to $\mathbb{R}^2$. Thus, if $\mathbf{x}: U \rightarrow V$ is a parametrization of a coordinate neighborhood of some $p \in S$, then, noting that $\mathbf{x}$ is differentiable on U and π is differentiable on $\mathbf{x}(U) = V$ (in the $\mathbb{R}^3 \rightarrow \mathbb{R}^2$ sense), $\pi \circ \mathbf{x}: U \rightarrow \mathbb{R}^2$ is differentiable on all of U , hence π is differentiable. $\square$

Problem. 2.3.4. Construct a diffeomorphism between the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

and the sphere $x^2 + y^2 + z^2 = 1$.

Solution. Let E be the ellipsoid and S^2 the sphere. Then, consider the map $f: E \rightarrow S^2$ given by $f(x, y, z) = (x/a, y/b, z/c)$. Clearly this is a bijection between points on E and S^2 , for $f(x, y, z)$ is in S^2 iff

$$(x/a)^2 + (y/b)^2 + (z/c)^2 = 1$$

which holds iff (x, y, z) is on E , and this map has an inverse $f^{-1}(x, y, z) = (ax, by, cz)$. Moreover, f is differentiable because it is the restriction of a differentiable function $\mathbb{R}^3 \rightarrow \mathbb{R}^3$ to E and its inverse is differentiable because it is similarly the restriction of a differentiable function $\mathbb{R}^3 \rightarrow \mathbb{R}^3$ to S^2 . Thus f is a diffeomorphism between E and S^2 . $\square$

Problem. 2.3.5. Let $S \subseteq \mathbb{R}^3$ be a regular surface, and let $d: S \rightarrow \mathbb{R}$ be given by $d(p) = |p - p_0|$, where $p \in S$, $p_0 \in \mathbb{R}^3$, $p_0 \notin S$. Prove that d is differentiable.

Solution. We show that $d': \mathbb{R}^3 \rightarrow \mathbb{R}$ defined by $d'(p) = |p - p_0|$ (so $d'|_S = d$) is differentiable at all points besides p_0 , at which point the result follows because the restriction of a map $\mathbb{R}^3 \rightarrow \mathbb{R}$ that is differentiable at x to a map $S \rightarrow \mathbb{R}$ is still differentiable at x .

We first show that $d'(p)^2$ is differentiable on all of $\mathbb{R}^3$; this follows from $d'(p)^2 = |p - p_0|^2 = (p - p_0) \cdot (p - p_0)$ and so its derivative is the linear map given by the 1×3 matrix $2(p - p_0)$ (so the derivative at a point x is $2(p - p_0) \cdot x$). Then, $\sqrt{d'(p)^2}$ is differentiable everywhere except at p_0 because $\sqrt{\bullet}: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ is differentiable on all positive reals but not at zero.

By the first paragraph, this finishes the problem. $\square$